

# Smartphone Usage Behavior Analytics Executive Summary

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# Smartphone Usage Behavior Analytics

## Executive Summary

Data Source: Kaggle Smartphone Usage & Addiction dataset, 7,500 users

This analysis examined smartphone usage patterns and addiction tendencies using a dataset of 7,500 users. The objective was to identify behavioral indicators associated with smartphone addiction and evaluate potential impacts on sleep, stress, and academic or work performance. The dataset included demographic information, usage behavior, lifestyle factors, device interaction metrics, and addiction classifications.

The analysis found that smartphone dependency is widespread, with more than 70% of users classified as addicted and nearly 90% exhibiting some level of addiction. Moderate addiction levels were more common than severe addiction levels, suggesting many users demonstrate problematic usage behaviors before reaching the highest risk category. Users classified as addicted averaged significantly higher weekend screen time than non-addicted users.

Screen time emerged as the strongest indicator of smartphone dependency. Average daily screen time exceeded about eight hours, while weekend screen time averaged more than nine hours and reached as high as 14.80 hours for some users. Screen time was consistently higher than time spent on social media, gaming, or work and study activities, indicating that smartphone engagement extends beyond traditional activity categories.

The analysis also identified relationships between smartphone usage, sleep patterns, and stress levels. More than half of all users reported six hours or less of sleep, while higher stress levels were associated with reduced sleep duration. Although addiction levels showed a relationship with reported impacts on academic and work performance, the distribution of impact closely mirrored overall addiction rates, suggesting smartphone usage may affect productivity across all levels of dependency.

A limitation of the dataset was the lack of detailed screen time categorization. Additional information regarding application-specific usage, messaging activity, media consumption, shopping behavior, and other smartphone activities would provide greater insight into the behavioral drivers of addiction and support more targeted recommendations.

Based on these findings, individuals concerned about excessive smartphone usage should consider implementing screen time limits, reducing device accessibility during personal time, and increasing participation in offline activities that promote healthier work-life balance and social engagement.

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# 1. Analysis, Findings, and Insights

## 1.1 Detailed Findings and Analysis

### 1.1.1 Methodology

- The data was provided at Kaggle.com, Smartphone Usage and Analysis 7500 Rows in csv format. Excel was used for initial glance of data and cleaned in MySQL Workbench. It was explored and modified with Microsoft Power BI Query where visualizations were developed and the dashboard was created.

### 1.1.2 Statistics

- The data consisted of 7,500 users with 2,461 identified as female, 2,553 male and 2,486 identified as Other.
- The users' age ranged from 18 to 35 with 27 being the average age.
- The range of daily screen time hours stands out at 3.5 hours. The other range of hours per day are within 1 to 1.5 hours of each other.
- The mean and mode are very close within about a minute of each other in all categories of daily usage hours.
- Stress levels are very close for all users. The difference between low, medium and high levels are about 61 users.
- The mean\median hours of sleep are 6 hours 44 minutes / 6 hours 43 minutes.
- Daily addiction levels: none 10.92%, mild 18.31%, moderate 38.32% and severe 32.45%.

See figure 1 on page 4.

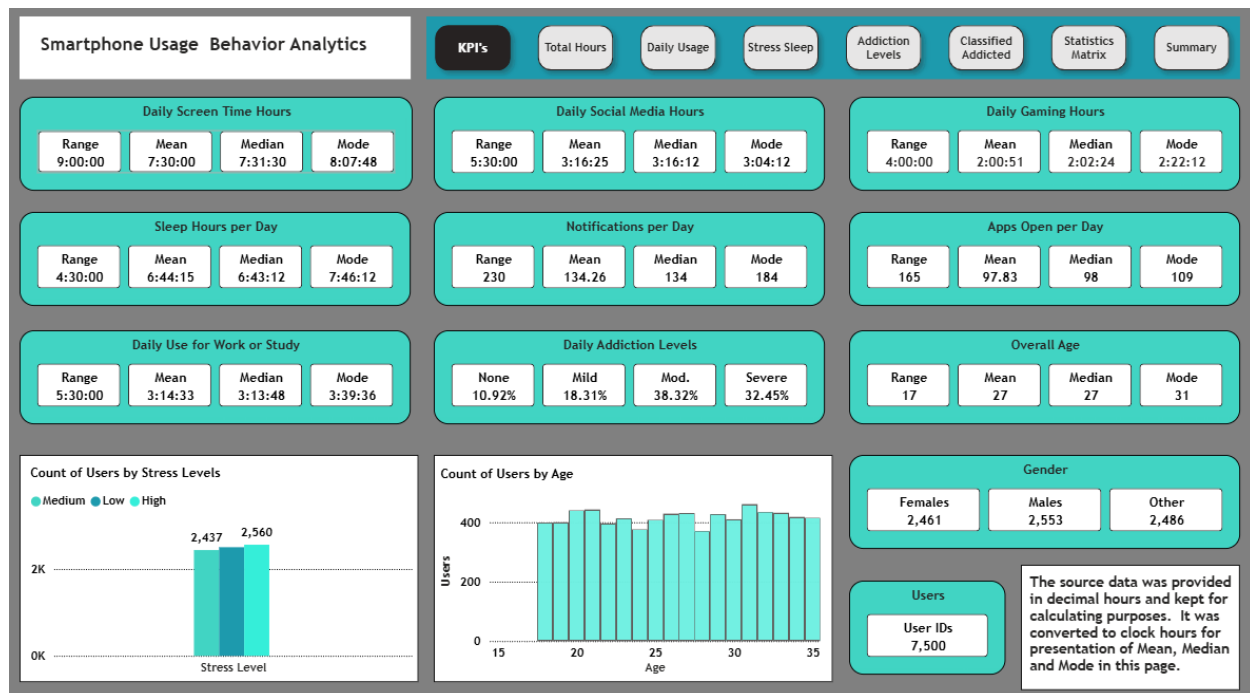


Figure 1. Overview

### 1.1.3 Usage Behavior

- The distributions of unfiltered data of different hours of usage by the number of users, is even across the column charts with no outliers or major deviations, within each usage category.
- However, the screen time and weekend screen time categories separate themselves from the rest of the categories.
- The screen time hours and weekend screen time hours are significantly higher than other usage activity. Weekend screen time hours range from 3.5 to 14.80. Daily screen time hours range from 3 to 12 hours.
- When filtering categories by gender and age there are no outliers nor much variance across the columns.
- The genders are in close range for each group. Although the males have slight lead in each usage area.
- The males spend more time on their smartphone in each category cumulatively by age of the 3 genders combined by 1,241.48 decimal hours. Screen time 677.34, social media 328.69, Gaming 129.91 and work / study 105.54.
- The average age is 27 dominated by the males.

See figure 2 on page 5.



**Figure 2. Screen Time**

## 1.1.4 Lifestyle Factors

- Smartphones do influence sleep. Up to 56% of all users receive 6 or less hours of sleep. The gender of the other leads this group with 58%.
- Stress also has an influence on sleep. Sleep by stress levels is as low as 4.5 hours and high as 9 hours sleep for each level. But further analysis revealed the data 9 hours only represents 1 or 2 people. The most frequent hours of sleep with high stress were 5.58 hours for 14 people while 2,560 users had high stress.
- Medium stress reflected the most frequent hour of sleep was 8.96 for 12 users. While 2,503 users had medium stress.
- Looking at average usage of weekend screen time is almost 3 times higher than social media, gaming and work study. Screen time is almost twice as high as the other usage categories.
- There is a correlation between the average chart and maximum chart. In the maximum chart the hours go up significantly. Social media, gaming and work study double in height. Screen time and weekend screen time increases about 5 hours higher on the maximum chart. However, the patterns in each chart closely mirror each other.

- The screen time data provided was not broken down into further categories for a better data analysis of usage habits and tendencies contributing to addiction.
- Screen time is not the sum of all other categories collectively. It is also normal for screen time not to equal the total of individual categories when analyzing screen hours. Screen time includes music, movies, shopping, texting, news apps, photos, Kindle, shopping online and banking. There is a plethora of apps to choose from in the app store.
- Users can use multiple apps or device at the same time. They can multitask doing activities with many apps simultaneously which can overlap categories, tracking systems not reporting accurate categories. A student can have a video or music playing at the same time texting a friend or be on social media. The tracking system could track 45 minutes with video playing and 45 minutes of texting for a total of 90 minutes. The true time would only be 45 minutes. Some users may put their phone down while an app is open or leave it sit while its updates, recording background activity verse active screen time looking at their device.

### 1.1.5 Device Interaction

- Users on average are opening 97.83% of apps per day compared to an average of 134.26 notifications per day. That equates to 72.86% of the number of notifications.
- The average apps open per day where not classified as addicted is even with the average apps open by those classified as addicted.

### 1.1.6 Addiction Metrics

- Over 70% of the users are classified as addicted.
- Weekend screen time hours and screen time hours deviate from the norm with addicted and non-addicted users. Although the bar chart shows addicted users with a total of 54,215.88 hours and non-addicted users with 15,112.82 hours, their patterns are very close and similar.
- Overall moderate addiction level is higher than severe level. There are 440 moderate users higher than the severe users at 5.87%.

See figure 3 on page 7.



Figure 3. Addiction

### 1.1.7 Impact Indicators

- There are 89.08% of users with some level of addiction and 10.02% with no addiction level.
- There is a correlation between addiction level and work study impact level. Where academics or work is impacted, the impacted level percent by addiction level is within 1% of the corresponding addiction level percent.
- The trajectory of addiction levels with each gender has the same pattern when stress is high. Moderate stress continues to be higher than severe stress.
- Males lead in the high stress level by 5 or 0.51%.
- Users on average are opening 97.83% apps per day compared to an average of 134.26 notifications per day. That equates to 72.86% of the number of notifications.
- The average apps open per day where not classified as addicted is even with the average apps open by those classified as addicted.
- The Matrix chart reveals those classified as addicted have 3.30 hours more screen time than those not addicted. They have 1.5 more hours on social media than those not addicted.